

The stability of solitary waves

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The stability of finite amplitude, two-dimensional solitary waves of permanent form in water of uniform depth with respect to two-dimensional infinitesimal disturbances is investigated. It is numerically confirmed that the recent analytical results obtained by Saffman (submitted to *J. Fluid Mech.*) for the "superharmonic" instability of periodic waves hold also in the case of solitary waves.

I. INTRODUCTION

In this paper, we study the stability of finite amplitude, two-dimensional solitary waves of permanent form in water of uniform depth with respect to two-dimensional infinitesimal disturbances. The stability of solitary waves was investigated by Jeffrey and Kakutani¹ on the basis of the Korteweg-de Vries (KdV) equation. They showed that the corresponding eigenvalue problem possesses no eigenmode besides the trivial neutral one corresponding to the pure phase shift of the basic wave, and that the solitary wave solution of the KdV equation is neutrally stable with respect to any infinitesimal disturbance that vanishes at infinity. However, the validity of their analysis is of course restricted to the region of very small amplitude where the KdV equation holds, and the stability problem of the solitary wave of an arbitrary amplitude still remains unresolved.

On the other hand, the stability problem of periodic waves has been investigated extensively since Longuet-Higgins^{2,3} first considered this problem. In a previous work,⁴ we studied the stability of periodic waves of permanent form in deep water with respect to "superharmonic" disturbances and found numerically that an exchange of stability occurs at the steepness ak , where the total energy and the impulse of the wave first become stationary as functions of ak . However, as shown by Longuet-Higgins,⁵ the total energy and the impulse of periodic waves always become stationary at the same values of ak when the water is infinitely deep. Therefore we could not decide at that time which quantity of the two was of decisive importance for this type of instability.

We planned to investigate the stability of solitary waves because of the following considerations. For solitary waves, all the disturbances are superharmonic in a sense, so their stability characteristics would probably be closely related to those of periodic waves for superharmonic disturbances. Moreover, for solitary waves, the total energy and the impulse reach their extrema at distinct amplitudes. By investigating the case of solitary waves, we hoped that we would be able to decide which of the two quantities (the total energy or the impulse) was of critical importance for the superharmonic instability of periodic waves. This was the original motivation of this work.

Quite recently, however, Saffman⁶ proved analytically that, for periodic waves, an exchange of stability occurs for superharmonic disturbances at every stationary point of the

total energy. From this, it has become clear that the quantity of critical importance for the superharmonic instability of periodic waves is not the impulse but the total energy. Though the question that originally motivated this work has thus been resolved, nevertheless we believe that it is still worthwhile to examine numerically the extent to which Saffman's analytical results for periodic waves hold in the case of solitary waves.

Aside from its relevance to the stability of periodic waves mentioned above, the stability of solitary waves also has its own importance, of course. As discussed by Peregrine,⁷ for waves approaching a sloping beach, the interaction between neighboring waves is usually so weak that it is appropriate to treat them as solitary waves instead of periodic waves. Because of this fact, the stability characteristics of solitary waves would have great importance in several problems concerning the shoaling waves, such as determining the position where the incident waves begin to break.

In Sec. II, the solitary wave solutions of the full water wave equations are calculated. The stability problem of these waves is formulated in Sec. III and the numerical results are discussed in Sec. IV.

II. SOLITARY WAVE SOLUTION

We consider two-dimensional, inviscid, irrotational solitary waves of permanent form in water of uniform depth. The flow field can be reduced to steady state by taking a frame of reference moving horizontally with the phase speed c of the wave. All quantities are assumed to be nondimensionalized, with the undisturbed depth h as the unit length and the phase speed c of the wave as the unit speed. We introduce the complex velocity potential $W (= \Phi + i\Psi)$, choosing $\Phi = 0$ at the crest and $\Psi = 0$ at the flat bottom. The fluid region is then mapped onto the strip $0 < \Psi < 1$, $-\infty < \Phi < \infty$ on the W plane with $\Psi = 1$ corresponding to the free surface.

Let us introduce a quantity Ω defined by $\Omega = \ln(dW/dz)$, where $z = x + iy$, with x being the horizontal coordinate and y the vertical one pointing upward. In terms of the magnitude q of the velocity and the angle ϑ between the velocity and the x axis, Ω can also be expressed as $\Omega = \tau - i\vartheta$ with $\tau = \ln q$. Throughout the fluid region, Ω is an analytic function of z and thus of W , and tends to zero at infinity.

There are two boundary conditions that should be satis-

fied by the flow field, i.e., the Bernoulli condition at the free surface and the kinematic condition at the bottom. These conditions can be expressed as

$$\frac{dq^3}{d\Phi} = -\frac{3}{F^2} \sin \vartheta, \quad \text{on } \Psi = 1 \quad (1)$$

and

$$\vartheta = 0, \quad \text{on } \Psi = 0, \quad (2)$$

where F is the Froude number defined by $F = c/\sqrt{gh}$.

The problem of solving a solitary wave solution can thus be transformed into the problem of finding a complex function Ω that is analytic with respect to W within the region of the unit strip $0 < \Psi < 1$, decays at infinity, and satisfies the two boundary conditions (1) and (2).

By applying Cauchy's integral theorem, we can find the following relation between the values of τ and ϑ on the free surface $\Psi = 1$, i.e.,

$$\begin{aligned} -\vartheta(\Phi) - \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\vartheta(\varphi)}{(\varphi - \Phi)^2 + 4} d\varphi \\ = -\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{(\varphi - \Phi)\tau(\varphi)}{(\varphi - \Phi)^2 + 4} d\varphi \\ + \frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{\tau(\varphi)}{\varphi - \Phi} d\varphi, \end{aligned} \quad (3)$$

where the last integral is of Cauchy's principal-value form. The functions $\vartheta(\Phi)$ and $\tau(\Phi)$ denote $\vartheta(\Phi, \Psi = 1)$ and $\tau(\Phi, \Psi = 1)$, respectively.

As a parameter for measuring the amplitude of a solitary wave, we will employ the nondimensional fluid velocity q_c at the crest. Generally q_c lies between 0 and 1. When $q_c = 0$, the crest is a stagnation point and the wave is of limiting height with a 120° angular crest. On the other hand, q_c close to 1 corresponds to infinitesimal waves.

For a given value of q_c , the corresponding solitary wave solution can be obtained by an iterative procedure as follows. First, some appropriate functional form is assumed for $\tau(\Phi)$, which satisfies the conditions $\tau(0) = \ln q_c$ and $\tau(\infty) = 0$. Then, the corresponding $\vartheta(\Phi)$ can be obtained from the integral equation (3). On the other hand, condition (1) implies

$$q^3(\Phi) - q_c^3 = -\frac{3}{F^2} \int_0^\Phi \sin \vartheta(\varphi) d\varphi, \quad (4)$$

$$1 - q_c^3 = -\frac{3}{F^2} \int_0^\infty \sin \vartheta(\varphi) d\varphi, \quad (5)$$

especially for $\Phi \rightarrow \infty$ (note that $q \rightarrow 1$ as $|\Phi| \rightarrow \infty$).

The Froude number F can be determined from $\vartheta(\Phi)$ by using Eq. (5). Substituting this value for F into (4), we can evaluate $q(\Phi)$. From this $q(\Phi)$, a new estimation for $\tau(\Phi)$ can be calculated by the relation $\tau(\Phi) = \ln q(\Phi)$, thus completing one cycle of iteration. By repeating this cycle until sufficient convergence is achieved, we can obtain a solitary wave solution with an arbitrary accuracy.

For the calculation of steep solitary waves, it is convenient to introduce a new independent variable γ instead of Φ by the relation

$$\Phi = \alpha\gamma + \gamma^m, \quad (6)$$

where α is some small positive number and m is some posi-

tive odd integer. This variable transformation is similar to that of the Hunter and Vanden-Broeck.⁸ (Their transformation corresponds to the case of $\alpha = 0$.) The role of their transformation seems to be twofold, i.e., stretching the region near the steep crest and at the same time bringing the far region closer to the origin. In our transformation (6), these two roles are assigned separately to two terms; the role of stretching the region near the crest is assigned to the term $\alpha\gamma$ (with $\alpha \ll 1$) instead of γ^m . By making use of this new degree of freedom, we can choose more appropriate variables for the problem. There is another reason for the introduction of the additional term $\alpha\gamma$. In the next section, we will study the stability problem of solitary waves. In the formulation of the problem, an operator $d/d\Phi$ appears. If the term $\alpha\gamma$ were not present in the variable transformation (6), the operator $d/d\Phi$ would become singular at the origin when it was rewritten in terms of the new variable γ , and extra numerical manipulation would thus become necessary in the neighborhood of the origin.

Next, in order to solve these equations numerically, we introduce the N mesh points $\gamma_i = (i-1)\Delta$ ($i = 1, \dots, N$), with Δ the interval of mesh points. To evaluate the first two integrals in (3), we applied the usual trapezoidal rule. As to the integral on the rhs of (4), the eight-point Lagrangian integration formula was applied. In the evaluation of the principal-value integral of (3) at the mesh point γ_j , we introduced new mesh points $\tilde{\gamma}_i = (i-1/2)\Delta$ ($i = 1, \dots, N$), expressed the values of the integrand at these new mesh points in terms of those at the original ones by using the eight-point Lagrangian interpolation formula, and then applied the trapezoidal rule to them. In all the infinite integrals, the contributions from the region $\Phi > \Phi_{\max}$ with $\Phi_{\max} \equiv \alpha\gamma_N + \gamma_N^m$ were simply neglected.

In the actual calculations, the iteration was terminated when the difference in the values of F^2 for two successive iteration cycles was less than 10^{-7} ; this usually took about 20 iterations. For a given q_c , we repeated calculations for various values of α , m , N , and $\Phi_{\max}$, and if the result became independent of these four parameters, we employed it as the solution for that value of q_c . As a typical example of this process, we present some numerical data for the case $q_c = 0.11$ in Table I. The values of α and m are set equal to 0.0125 and 5, respectively, which appeared to be the most appropriate for this value of q_c .

In order to evaluate the amount of error involved in the solution thus obtained, the Bernoulli condition was checked; we calculated the value of $|(1/2q^2 + \eta/F^2) - (1/2q_c^2$

TABLE I. Values of F^2 for $q_c = 0.11$ ($\omega = 0.97982$) (α and m are set equal to 0.0125 and 5, respectively).

$\Phi_{\max} = 30$		$\Phi_{\max} = 40$	
N	F^2	N	F^2
30	1.6679349	30	1.6679344
40	1.6679357	40	1.6679356
50	1.6679358	50	1.6679357
60	1.6679358	60	1.6679358

TABLE II. Values of ω , I , E , and F^2 for various values of q_c .

q_c	ω	I	E	F^2
0.325	0.82473	2.62234	0.98979	1.65934
0.300	0.85002	2.62436	1.00057	1.66647
0.275	0.87361	2.62008	1.00594	1.67132
0.250	0.89537	2.61089	1.00661	1.67406
0.225	0.91520	2.59841	1.00356	1.67498
0.200	0.93302	2.58438	0.99795	1.67445
0.175	0.94877	2.57060	0.99106	1.67293
0.150	0.96240	2.55865	0.98415	1.67093
0.125	0.97392	2.54974	0.97834	1.66894
0.100	0.98333	2.54448	0.97439	1.66739
0.080	0.98933	2.54275	0.97275	1.66663
0.060	0.99400	2.54261	0.97228	1.66632
0.040	0.99733	2.54315	0.97246	1.66632

$+\eta(0)/F^2|$ at every mesh point, with η being the displacement of the free surface. This value usually remained less than $O(10^{-8})$ for any value of q_c considered as long as the number of mesh points was set larger than about 60.

Table II lists some of the calculated values of F^2 , the total energy E , and the impulse I of the wave; the values of ω defined by $\omega = 1 - F^2 q_c^2$ are also shown for the sake of comparison with other works. This table clearly shows that not only the first extrema (maxima) but also the second ones (first local minima) have successfully been detected for all three of the quantities F^2 , E , and I . The values of ω that correspond to the maxima of these quantities lie, respectively, within the following ranges:

$$0.8447 < \omega_I < 0.8451, \quad 0.8878 < \omega_E < 0.8881,$$

and

$$0.9166 < \omega_F < 0.9170.$$

Before concluding this section, we should refer to an unexpected but advantageous characteristic of our calculation scheme. In the next section, the stability problem is ultimately reduced to an eigenvalue problem of some coefficient matrix. The dimension of the matrices that should be handled is $2N - 1$, with N being the number of mesh points introduced. Therefore the stability calculation would be practical only when we can describe the solitary waves sufficiently accurately with only a small number of mesh points, even if the waves become steep. Table 2 of Hunter and Vanden-Broeck shows that about 150 mesh points were necessary in their calculation in order to obtain the value of the Froude number F , which is correct to five decimal places at the amplitude $\omega = 0.98$. In our calculations, the Froude number already converges up to six decimal places with as few as 40 mesh points, which is less than one-third of 150 for almost the same amplitude ($q_c = 0.11$; $\omega = 0.97982$), as shown in Table I. This characteristic of our formulation, which is probably attributable to the new transformation (6), will be a great advantage when we embark on the stability problem in the next section.

III. STABILITY PROBLEM

In this section we consider small disturbances superposed on a steady solitary wave. The displacements of the

free surface $\eta(x, t)$ and the velocity potential $\varphi(x, y, t)$ are generally expressed as

$$\eta(x, t) = H(x) + \tilde{\eta}(x, t),$$

$$\varphi(x, y, t) = \Phi(x, y) + \tilde{\varphi}(x, y, t),$$

where $H(x)$ and $\Phi(x, y)$ denote, respectively, the profile of the free surface and the velocity potential of the steady undisturbed wave, and $\tilde{\eta}$ and $\tilde{\varphi}$ are small disturbances of these quantities.

Linearizing the usual free-surface boundary conditions with respect to the small quantities with a tilde, we obtain the following linearized disturbance equations, which should be satisfied at the undisturbed free surface $y = H(x)$ or $\Psi = 1$:

$$\lambda \tilde{\eta} = -q \frac{d\tilde{\eta}}{ds} - \frac{1}{\cos \vartheta} \frac{d\tilde{\psi}}{ds} - \frac{1}{\cos \vartheta} \frac{d}{ds} (q \cos \vartheta) \tilde{\eta}, \quad (7)$$

$$\lambda \tilde{\varphi} = -q \frac{d\tilde{\varphi}}{ds} - \left(q \frac{d}{ds} (q \sin \vartheta) + \frac{1}{F^2} \right) \tilde{\eta},$$

where s denotes the arc length measured along the undisturbed free surface on the physical z plane with $s = 0$ at the crest, and $\tilde{\psi}$ is the stream function of the disturbance motion. In these equations, the time dependence of the disturbance has already been assumed as $e^{\lambda t}$. Therefore the functions $\tilde{\eta}$ and $\tilde{\varphi}$ are no longer arbitrary but instead are the eigenfunctions of the linearized problem.

Throughout this work, we are exclusively concerned with those eigenmodes that vanish at infinity. These eigenmodes are not sufficient to construct a spatially localized disturbance of an arbitrary form, and another family of eigenmodes that satisfies a weaker boundary condition, i.e., those that need not vanish at infinity but remain bounded, must be included. These "improper" eigenmodes, however, correspond to nothing but the usual monochromatic linear surface gravity waves superposed on a uniform flow at infinity and are, of course, neutrally stable. From this consideration, we can safely restrict our attention to those eigenmodes that vanish at infinity only insofar as the stability characteristics are concerned.

By exploiting the symmetry of the undisturbed wave about the crest, it can easily be shown from (7) that if $[\lambda, \tilde{\eta}(s), \tilde{\varphi}(s)]$ is a pair of eigenvalue and corresponding eigenfunctions, then so is $[-\lambda, -\tilde{\eta}(-s), \tilde{\varphi}(-s)]$. This immediately implies that the eigenvalues always appear in pairs $(\lambda, -\lambda)$ if they are real or pure imaginary, and in quartets $(\lambda, \lambda^*, -\lambda, -\lambda^*)$ if they are complex, with the asterisk denoting complex conjugation. As in the case of periodic waves, this fact also permits us to transform the eigenvalue problem (7) for λ to the following one for λ^2 :

$$\begin{aligned} \lambda^2 \tilde{\eta}^{(e)} = & \left(q \frac{d}{ds} + a \right)^2 \tilde{\eta}^{(e)} + \left(q \frac{d}{ds} + a \right) \frac{1}{\cos \vartheta} \frac{d}{ds} G(\tilde{\varphi}^{(e)}) \\ & + \frac{1}{\cos \vartheta} \frac{d}{ds} G \left(q \frac{d}{ds} \tilde{\varphi}^{(e)} + b \tilde{\eta}^{(e)} \right), \\ \lambda^2 \tilde{\varphi}^{(e)} = & q \frac{d}{ds} \left(q \frac{d}{ds} \tilde{\varphi}^{(e)} + b \tilde{\eta}^{(e)} \right) + b \left(q \frac{d}{ds} + a \right) \tilde{\eta}^{(e)} \\ & + \frac{b}{\cos \vartheta} \frac{d}{ds} G(\tilde{\varphi}^{(e)}), \end{aligned} \quad (8)$$

where

$$a \equiv \frac{1}{\cos \vartheta} \frac{d}{ds} (q \cos \vartheta), \quad b \equiv q \frac{d}{ds} (q \sin \vartheta) + \frac{1}{F^2},$$

$$\tilde{\eta}^{(o)} \equiv \frac{1}{2} \{ \tilde{\eta}(s) - \tilde{\eta}(-s) \}, \quad \tilde{\varphi}^{(e)} \equiv \frac{1}{2} \{ \tilde{\varphi}(s) + \tilde{\varphi}(-s) \}.$$

In these equations, G stands for an operator that solves the integral equation (3); that is, $-\vartheta = G(\tau)$. It should also be noted that the extent of the space variable has been reduced from $(-\infty, \infty)$ to $[0, \infty)$ by this transformation of the problem.

By requiring (8) to be satisfied at N equally spaced mesh points γ_i ($i = 1, \dots, N$) (introduced before), we obtain an eigenvalue problem for λ^2 with a coefficient matrix, e.g., C , with a dimension of $2N - 1$ as follows:

$$\lambda^2 \mathbf{x} = C\mathbf{x},$$

with

$$\mathbf{x} = (\tilde{\eta}_2^{(o)}, \tilde{\eta}_3^{(o)}, \dots, \tilde{\eta}_N^{(o)}; \tilde{\varphi}_1^{(e)}, \tilde{\varphi}_2^{(e)}, \dots, \tilde{\varphi}_N^{(e)}). \quad (9)$$

(Here $\tilde{\eta}_1^{(o)}$ has been omitted from the system of equations because it always remains zero by the definition of $\tilde{\eta}^{(o)}$.) In the reduction of (8) to (9), the derivative d/ds is first replaced by $q d\gamma/d\Phi \cdot d/d\gamma$, and a nine-point finite difference formula is then applied to the derivative with respect to γ .

Thus we have approximated the "original" eigenvalue problem (7), which is continuous in the space variable, by an eigenvalue problem (9) of a finite-dimensional matrix C . However, the relation between the eigenvalues of the "original" continuous problem and the approximate finite-dimensional one is very different in the present solitary wave case from the periodic wave case we investigated previously. In the case of the periodic waves, the original problem possessed an infinite number of eigenvalues. Therefore every eigenvalue of the approximate problem was a true one in the sense that it always had its own counterpart in the spectrum of the original problem as long as N was taken large enough for the calculation to be sufficiently accurate. On the other hand, in the present solitary wave case, it was anticipated that the original problem would not possess so many eigenvalues except when the amplitude of the basic wave approached very close to its limiting value. Therefore, even if we could obtain $2N - 1$ eigenvalues by solving Eq. (9), most of them would only be spurious and not correspond to those of the original problem. In order to pick up the true eigenvalues from these $2N - 1$ candidates, we repeatedly solved Eq. (9) for various choices of the parameters α , m , N , and $\Phi_{\max}$; if there were eigenvalues λ^2 of the matrix C that remained unchanged, we employed them as the true eigenvalues of the original problem.

IV. RESULTS AND DISCUSSIONS

The eigenvalue problem (9) was solved by the standard double-step QR algorithm.⁹ All the calculations were performed in double precision.

The eigenvalues we detected are plotted in Fig. 1 as functions of ω . Table III gives a rough estimate of the degree of convergence in the eigenvalues that we generally achieved. As shown in Fig. 1, we could not detect any eigenvalues besides the trivial zero at first. A new branch of eigen-

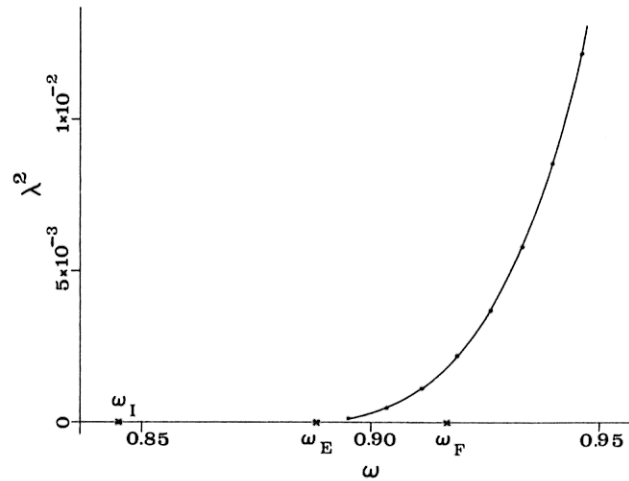


FIG. 1. The squares of the eigenvalues λ^2 at high values of the amplitude parameter ω .

value suddenly appeared, however, when we increased the amplitude, and it approached ω_E , i.e., the first extremum of the total energy.

In the case of periodic waves that was investigated previously, the branch of the eigenvalue that became unstable first intersected the axis $\lambda^2 = 0$ with a finite gradient at the critical amplitude. Therefore, we could locate the critical amplitude accurately by making use of an interpolation procedure without explicitly treating its close proximity. On the other hand, in the solitary wave case, the branch did not intersect the axis $\lambda^2 = 0$ but merged into it. We were obliged to treat the proximity of the critical amplitude if we wanted to decide its position precisely. In such a region, however, the relevant eigenvalue is necessarily very close to zero and it is impossible to calculate it with high accuracy because of the background numerical errors. [Table III shows that the calculated eigenvalues usually involve errors of $O(10^{-5})$, which is by no means negligibly small compared with the magnitude of the eigenvalue itself within the region of ω , e.g., between ω_E and 0.9.] For this reason, we cannot determine the critical amplitude precisely only by using the numerical data. Nevertheless, judging from the results shown in Fig. 1, it seems almost certain that the critical amplitude coincides with ω_E . Since the eigenvalue λ^2 we have detected above the critical amplitude is real and positive, one of the two eigenmodes corresponding to this grows exponentially in time. From this, it can be concluded that the solitary wave solution loses its stability at the amplitude that corresponds to the first extremum of the total energy.

TABLE III. Values of λ^2 for various choices of N and $\Phi_{\max}$ ($q_c = 0.22$, $m = 5$, and $\alpha = 0.06$).

	$\Phi_{\max} = 30$	$\Phi_{\max} = 40$	$\Phi_{\max} = 50$
$N = 50$	2.138×10^{-3}	2.144×10^{-3}	2.147×10^{-3}
$N = 60$	2.139×10^{-3}	2.144×10^{-3}	2.147×10^{-3}
$N = 70$	2.140×10^{-3}	2.145×10^{-3}	2.147×10^{-3}
$N = 80$	2.141×10^{-3}	2.145×10^{-3}	2.148×10^{-3}
$N = 90$	2.141×10^{-3}	2.146×10^{-3}	2.148×10^{-3}

If this instability actually has great relevance to the breaking of a solitary wave approaching a gently sloping beach, the breaker index (i.e., the depth-to-height ratio at the breaking point) would be probably around 1.2810.

There are many works by previous authors that have great relevance to the problem considered here, and the features of the numerical results we have just obtained can be reasonably understood in light of the results of these works.

Jeffrey and Kakutani¹ investigated the linear stability of the solitary wave solution of the KdV equation. By linearizing the KdV equation around its solitary wave solution, they obtained a linear eigenvalue problem that is the counterpart of our problem (7). They proved analytically that this eigenvalue problem possesses only one eigenmode under the boundary condition of vanishing at infinity, i.e., the trivial eigenmode that corresponds to the pure phase shift of the basic solitary wave. Because the eigenvalue for this trivial mode is of course zero, they reached the conclusion that the solitary wave solution of the KdV equation is neutrally stable. As the solitary wave solution of the full-water wave equations considered here asymptotically reduces to that of the KdV equation as its amplitude diminishes, its stability characteristics should also reduce to that of the solitary wave solution of the KdV equation. This implies that, under the boundary condition we have imposed at infinity, the eigenvalue problem (7) should possess no eigenvalue besides the trivial zero, at least within the region of very small amplitude, which is consistent with our numerical results.

The fact that the trivial zero is the only eigenvalue within the region of very small amplitude also has the following implication concerning the type of instability that would occur at larger amplitudes. In general, the eigenvalues depend continuously on the parameters involved in the problem. Therefore, in our case, the eigenvalues λ^2 should change continuously as the amplitude ω is gradually increased. In order not to violate this requirement, a new nontrivial branch of eigenvalue, if it ever existed, could emanate only from the axis $\lambda^2 = 0$. In other words, the type of instability that may possibly occur for the solitary wave is restricted to the so-called "exchange of stability" type. Although an instability that begins with a nonzero eigenvalue (i.e., $\text{Re } \lambda = 0$, while $\text{Im } \lambda \neq 0$ at the critical amplitude) does occur for "subharmonic" disturbances in the periodic wave case, the possibility of such an instability occurring is excluded in the case of solitary waves. In accordance with these considerations, the numerical results shown in Fig. 1 show that the instability actually begins with a zero eigenvalue. Moreover, they also show that it is not only at the critical amplitude that the unstable eigenvalue remains real even in the supercritical region, implying that the growth of the unstable eigenmode is monotonic and not oscillatory in time.

In the present work, we have restricted our attention to the eigenmodes that vanish at infinity as discussed before. In the actual calculation, this condition was imposed in various manners. First, in transforming the eigenvalue problem (7) to the reduced one (8), we applied Cauchy's integral theorem to the analytic function $\tilde{w} \equiv \tilde{\varphi} + i\tilde{\psi}$ and required that $\tilde{\varphi}$ and $\tilde{\psi}$ satisfy an integral equation the same as that satisfied by τ and $-\vartheta$ [i.e., Eq. (3)]. For this theorem to be

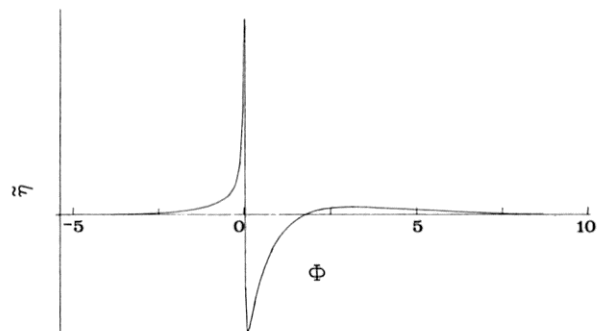


FIG. 2. A plot of an unstable eigenfunction $\tilde{\eta}$ as a function of Φ at $q_c = 0.18$ ($\omega = 0.9458$). The region of Φ considered in this calculation is $-40 < \Phi < 40$.

applicable, $\tilde{w}$ should vanish at infinity. Under this assumption, we approximated all the infinite integrals appearing in this integral equation by those with finite extent ($|\Phi| < \Phi_{\max}$) and neglected the contributions from the region $|\Phi| > \Phi_{\max}$. The condition at infinity was also taken into account in the construction of a matrix that stands for the differential operator d/ds appearing in (8). In order to show evidence that the unstable eigenmodes actually satisfy this condition and decay sufficiently rapidly at infinity, we show a plot of an unstable eigenfunction $\tilde{\eta}$ in Fig. 2.

As stated in the Introduction, this work was originally motivated in connection with the problem of superharmonic instability of periodic waves. In our previous work,⁴ we examined this problem numerically and found that an exchange of stability occurs at the first extrema of the total energy and the impulse (but not of the phase speed). However, this result also seemed to imply the existence of superharmonic bifurcation, the possibility of which had already been denied by Chen and Saffman's calculation.¹⁰ Inspired by this paradox, Longuet-Higgins¹¹ and the author¹² performed intensive calculations again and succeeded in resolving the apparent contradiction between the nonexistence of superharmonic bifurcation and the existence of an exchange of stability. The contradiction was resolved by the fact that the unstable eigenmode becomes equivalent to the pure phase shift at the critical amplitude. With this finding, the work of Chen and Saffman and our previous work became compatible. The controversy concerning the superharmonic instability and bifurcation of periodic waves has finally been settled by the recent analytical work of Saffman.⁶ By making use of Zakharov's Hamiltonian formulation, he proved that the necessary and sufficient condition for superharmonic disturbances to lose their stability at an amplitude is that the total energy of the basic wave becomes stationary at the amplitude. He also showed analytically the equivalence of the unstable eigenmode and the pure phase shift at the critical amplitude.

Saffman's analytical proof was formally restricted to the superharmonic instability of periodic waves. To our understanding, however, it is equally valid in the case of solitary waves as well if we reinterpret the index n of his work, which distinguished each Fourier mode, as a continuous parameter instead of an integer. The numerical results we have obtained indicate that this is actually the case. As we have

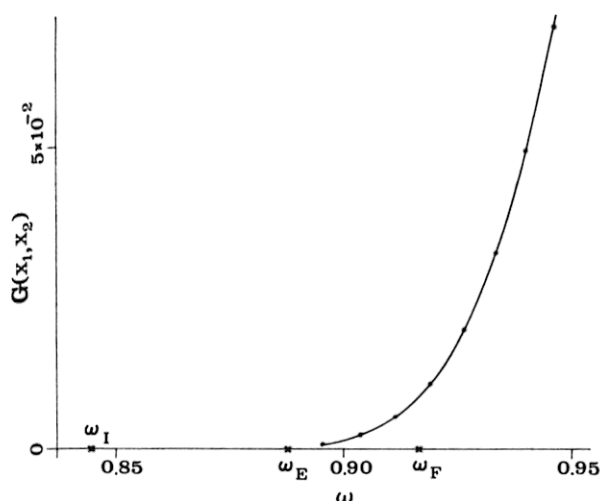


FIG. 3. The values of $G(\mathbf{x}_1, \mathbf{x}_2)$ as a function of the amplitude parameter ω .

already shown in Fig. 1, the exchange of stability occurs just at the extremum of the total energy in accordance with Saffman's result. In order to check another part of his results, i.e., the equivalence of two eigenmodes at the point of exchange of stability mentioned above, we calculated the quantity $G(\mathbf{x}_1, \mathbf{x}_2)$ defined by

$$G(\mathbf{x}_1, \mathbf{x}_2) = 1 - (\mathbf{x}_1, \mathbf{x}_2)^2 / |\mathbf{x}_1|^2 |\mathbf{x}_2|^2,$$

with

$$\mathbf{x}_1 = (\sin \vartheta_2, \sin \vartheta_3, \dots, \sin \vartheta_N)$$

and

$$\mathbf{x}_2 = (\tilde{\eta}_2^{(0)}, \tilde{\eta}_3^{(0)}, \dots, \tilde{\eta}_N^{(0)}),$$

i.e., $\mathbf{x}_1$ is the upper half of the eigenvector corresponding to the pure phase shift, while $\mathbf{x}_2$ is that of the mode that becomes neutral at the first extremum of the total energy. Here $G(\mathbf{x}_1, \mathbf{x}_2)$ measures the degree of linear independence of two vectors $\mathbf{x}_1$ and $\mathbf{x}_2$ and vanishes only when the two become linearly dependent. The value of $G(\mathbf{x}_1, \mathbf{x}_2)$ is plotted in Fig. 3 as a function of ω . It can be seen from this figure that the

linear independence of the two vectors gradually diminishes as the critical point is approached, and that the two vectors actually become linearly dependent just at the critical amplitude $\omega = \omega_E$.

Thus it has been confirmed numerically that the analytical results obtained by Saffman in the case of superharmonic instability of periodic waves remain valid also in the case of solitary waves. Accordingly there are several common features in the stability characteristics of the two cases as shown above. Some differences still exist of course. Among them, the most remarkable one may be the behavior of the unstable eigenvalue around the critical point. In the case of periodic waves, the eigenvalue depends on the amplitude as $\lambda^2 \propto \omega - \omega_E$ around the critical point. On the other hand, in the case of solitary waves, it behaves as $\lambda \propto \omega - \omega_E$ just above the critical amplitude. The theory of Saffman does not give any information about the parameter dependence of the unstable eigenvalues around the critical amplitude and we have not yet succeeded in obtaining any analytical explanation for this difference.

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